

Application

Applications to Computer Graphics

EXAMPLE 2 Given $A = \begin{bmatrix} 1 & .25 \\ 0 & 1 \end{bmatrix}$, describe the effect of the shear transformation $\mathbf{x} \mapsto A\mathbf{x}$ on the letter N in Example 1.

SOLUTION By definition of matrix multiplication, the columns of the product AD contain the images of the vertices of the letter N.

$$AD = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 0 & .5 & 2.105 & 6 & 8 & 7.5 & 5.895 & 2 \\ 0 & 0 & 6.420 & 0 & 8 & 8 & 1.580 & 8 \end{bmatrix}$$

The transformed vertices are plotted in Fig. 2, along with connecting line segments that correspond to those in the original figure.

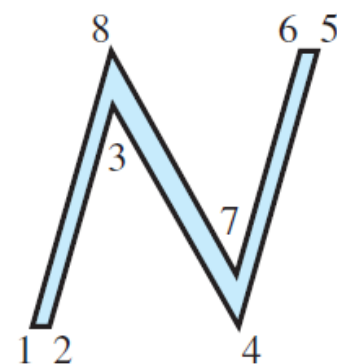


FIGURE 2
Slanted N .

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The italic N in Fig. 2 looks a bit too wide. To compensate, shrink the width by a scale transformation that affects the x -coordinates of the points.

EXAMPLE 3 Compute the matrix of the transformation that performs a shear transformation, as in Example 2, and then scales all x -coordinates by a factor of $.75$.

SOLUTION The matrix that multiplies the x -coordinate of a point by $.75$ is

$$S = \begin{bmatrix} .75 & 0 \\ 0 & 1 \end{bmatrix}$$

So the matrix of the composite transformation is

$$\begin{aligned} SA &= \begin{bmatrix} .75 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & .25 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} .75 & .1875 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

The result of this composite transformation is shown in Fig. 3.



FIGURE 3

Composite transformation of N .

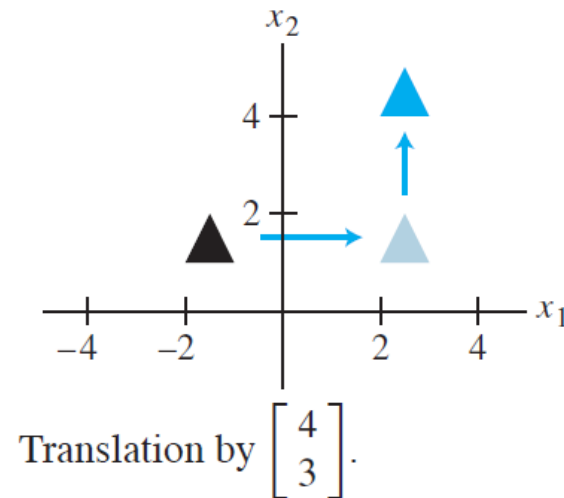
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Homogeneous Coordinates

The mathematics of computer graphics is intimately connected with matrix multiplication. Unfortunately, translating an object on a screen does not correspond directly to matrix multiplication because translation is not a linear transformation.

The standard way to avoid this difficulty is to introduce what are called *homogeneous coordinates*.

Homogeneous Coordinates refer to a mathematical representation where any two-dimensional or three-dimensional linear transformation can be expressed using a three-by-three or four-by-four matrix, respectively.



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Homogeneous Coordinates

Each point (x, y) in $\mathbb{R}^2$ can be identified with the point $(x, y, 1)$ on the plane in $\mathbb{R}^3$ that lies one unit above the xy -plane. We say that (x, y) has *homogeneous coordinates* $(x, y, 1)$. For instance, the point $(0, 0)$ has homogeneous coordinates $(0, 0, 1)$. Homogeneous coordinates for points are not added or multiplied by scalars, but they can be transformed via multiplication by 3×3 matrices.

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EXAMPLE 4 A translation of the form $(x, y) \mapsto (x + h, y + k)$ is written in homogeneous coordinates as $(x, y, 1) \mapsto (x + h, y + k, 1)$. This transformation can be computed via matrix multiplication:

$$\begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + h \\ y + k \\ 1 \end{bmatrix} \quad \blacksquare$$

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EXAMPLE 5 Any linear transformation on $\mathbb{R}^2$ is represented with respect to homogeneous coordinates by a partitioned matrix of the form $\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix}$, where A is a 2×2 matrix. Typical examples are

$$\begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

Counterclockwise
rotation about the
origin, angle φ

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

Reflection
through $y = x$

$$\begin{bmatrix} s & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

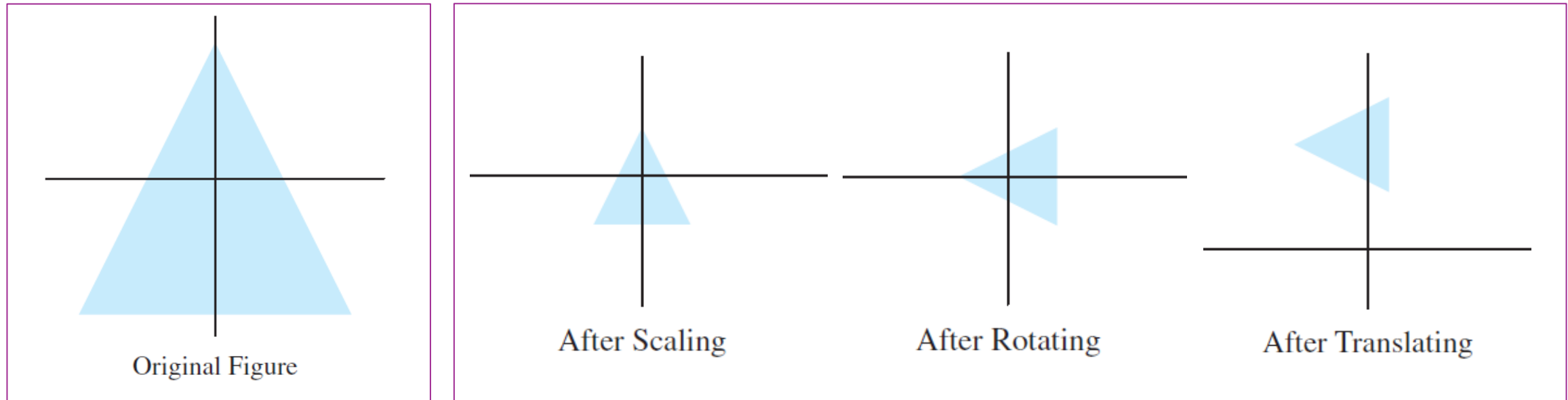
Scale x by s
and y by t



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Composite Transformations

The movement of a figure on a computer screen often requires two or more basic transformations. The composition of such transformations corresponds to matrix multiplication when homogeneous coordinates are used.



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EXAMPLE 6 Find the 3×3 matrix that corresponds to the composite transformation of a scaling by .3, a rotation of 90° about the origin, and finally a translation that adds $(-.5, 2)$ to each point of a figure.

SOLUTION If $\varphi = \pi/2$, then $\sin \varphi = 1$ and $\cos \varphi = 0$. From Examples 4 and 5, we have

$$\begin{aligned} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} &\xrightarrow{\text{Scale}} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &\xrightarrow{\text{Rotate}} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &\xrightarrow{\text{Translate}} \begin{bmatrix} 1 & 0 & -.5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \end{aligned}$$

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SOLUTION (Continued...)

The matrix for the composite transformation is

$$\begin{bmatrix} 1 & 0 & -.5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 0 & -1 & -.5 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -.3 & -.5 \\ .3 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \blacksquare$$

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EXAMPLE 7

Rotation of a figure about a point $\mathbf{p}$ in $\mathbb{R}^2$ is accomplished by first translating the figure by $-\mathbf{p}$, rotating about the origin, and then translating back by $\mathbf{p}$. See Fig. 7. Construct the 3×3 matrix that rotates points -30° about the point $(-2, 6)$, using homogeneous coordinates.

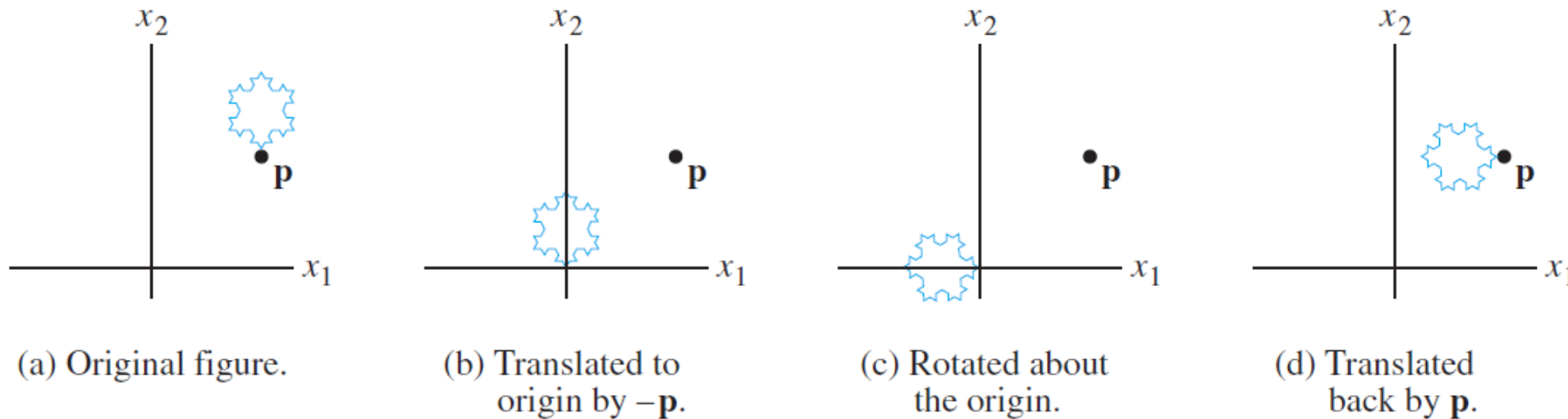


FIGURE 7 Rotation of figure about point $\mathbf{p}$.

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SOLUTION

Assemble the matrices right-to-left for the three operations. Using $\mathbf{p} = (-2, 6)$, $\cos(-30^\circ) = \sqrt{3}/2$, and $\sin(-30^\circ) = -.5$, we have

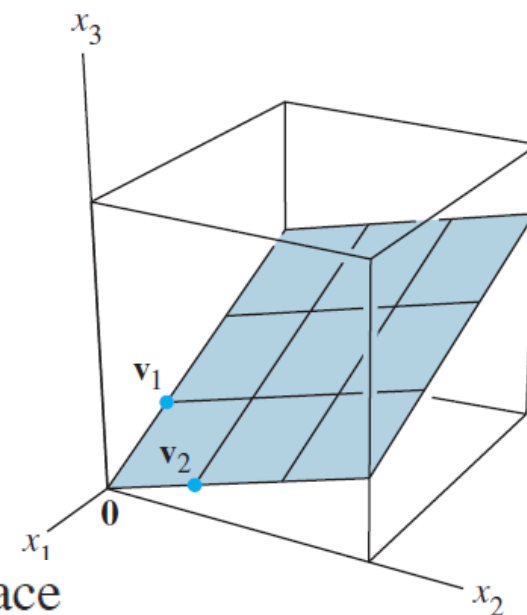
$$\begin{array}{ccc} \text{Translate} & \text{Rotate around} & \text{Translate} \\ \text{back by } p & \text{the origin} & \text{by } -p \\ \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix} \\ & = \begin{bmatrix} \sqrt{3}/2 & 1/2 & \sqrt{3} - 5 \\ -1/2 & \sqrt{3}/2 & -3\sqrt{3} + 5 \\ 0 & 0 & 1 \end{bmatrix} & \end{array}$$

Subspace of a Matrix

Subspace of $\mathbb{R}^n$

A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H .
- For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
- For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .



For instance, a plane through the origin is the standard way to visualize the subspace

Definition. A *subspace* of $\mathbb{R}^n$ is a subset V of $\mathbb{R}^n$ satisfying:

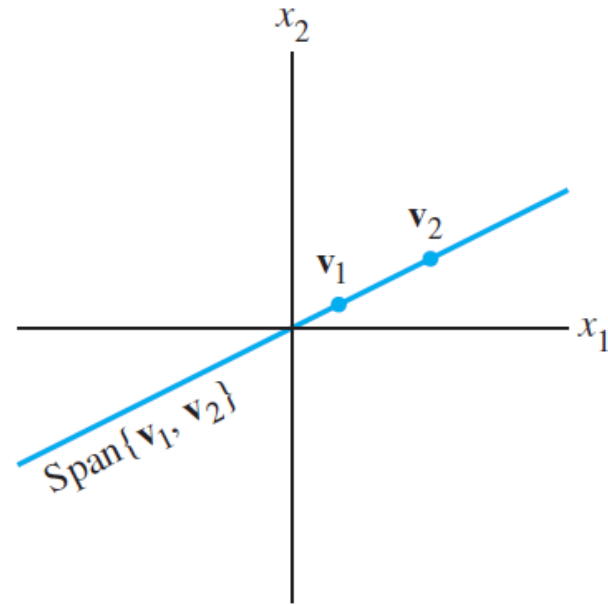
- Non-emptiness:** The zero vector is in V .
- Closure under addition:** If u and v are in V , then $u + v$ is also in V .
- Closure under scalar multiplication:** If v is in V and c is in $\mathbb{R}$, then cv is also in V .

Note that $\mathbb{R}^n$ is a subspace of itself because it has the three properties required for a subspace. Another special subspace is the set consisting of only the zero vector in $\mathbb{R}^n$. This set, called the **zero subspace**, also satisfies the conditions for a subspace.

Subspace of a Matrix

EXAMPLE 1

If $\mathbf{v}_1$ is not zero and if $\mathbf{v}_2$ is a multiple of $\mathbf{v}_1$, then $\mathbf{v}_1$ and $\mathbf{v}_2$ simply span a *line* through the origin. So a line through the origin is another example of a subspace.



$$\mathbf{v}_1 \neq \mathbf{0}, \mathbf{v}_2 = k\mathbf{v}_1.$$

Subspace of a Matrix

EXAMPLE 2 A line L *not* through the origin is *not* a subspace, because it does not contain the origin, as required. Also, Fig. 2 shows that L is not closed under addition or scalar multiplication. ■

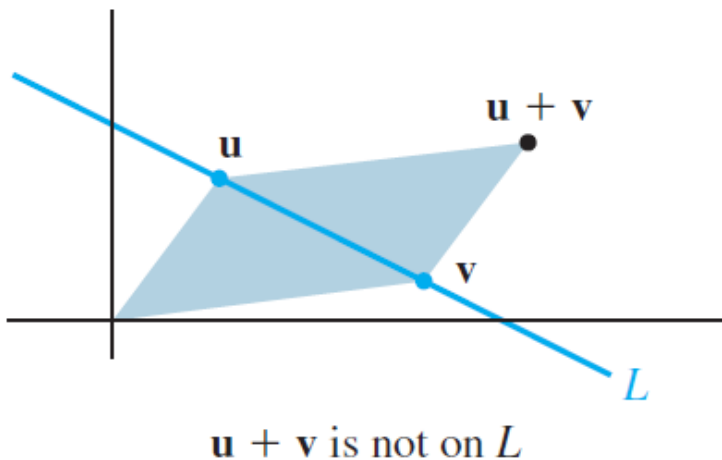
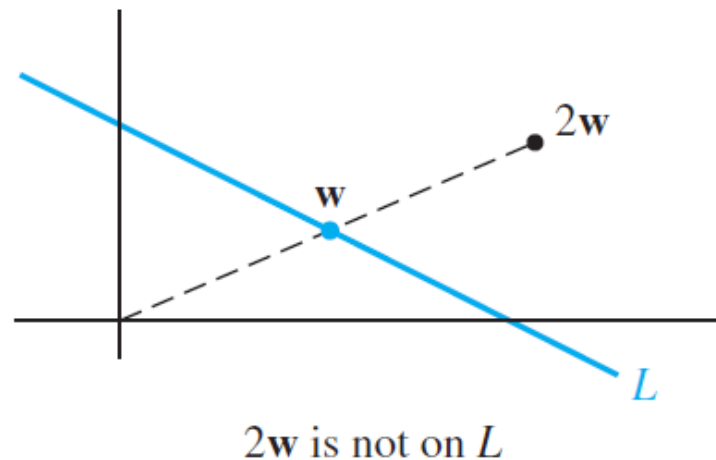


FIGURE 2



Example:- **predicting the price of a house based on its size** and other relevant features.

Scenario:

We want to model the relationship between house prices and their sizes. We will use a simple linear regression model, which is a linear model that predicts the dependent variable (house price) based on independent variables (house size in square feet, number of bedrooms, etc.).

Data:

Suppose we have the following dataset:

House Size (sq ft)	Number of Bedrooms	Price (\$)
1500	3	300,000
2000	4	400,000
2500	4	500,000
3000	5	600,000

We want to fit a linear model to predict the house price based on the house size and number of bedrooms.

Step 1: Formulate the Linear Model

We can represent the linear model as follows:

$$\text{Price} = w_1 \cdot \text{Size} + w_2 \cdot \text{Bedrooms} + b$$

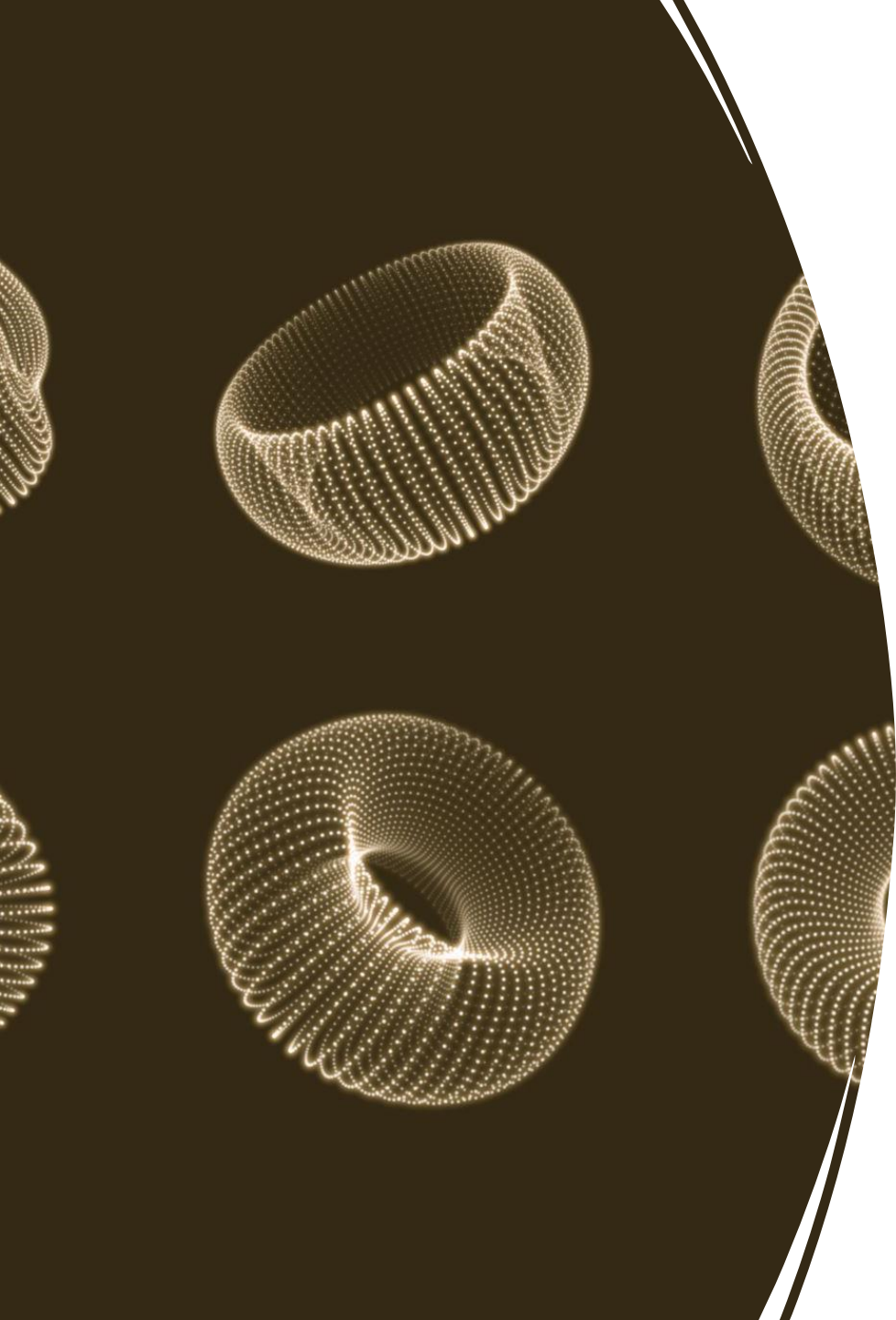
Where:

- w_1 is the weight (coefficient) for house size,
- w_2 is the weight for the number of bedrooms,
- b is the bias (intercept).

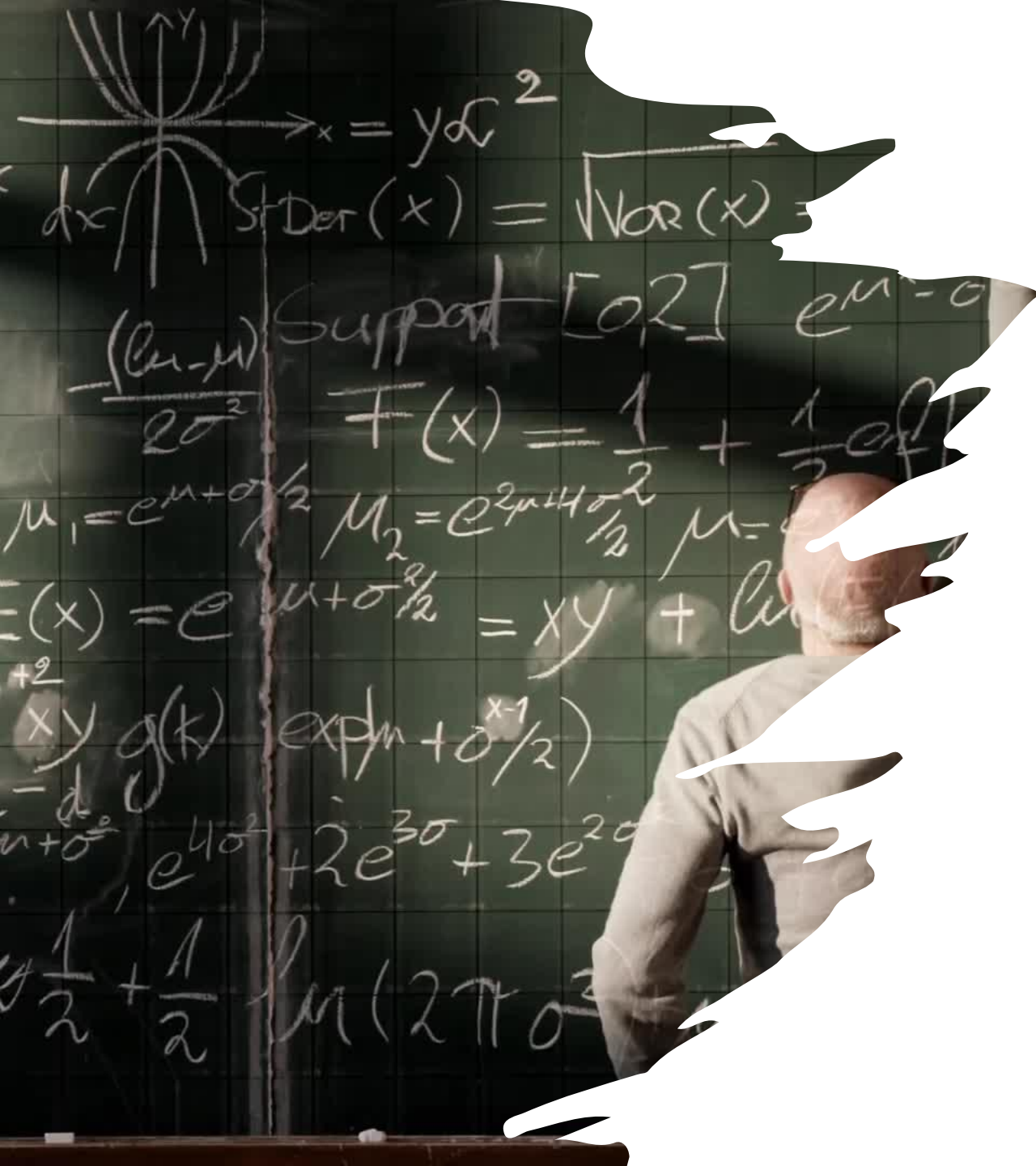
Step 2: Create Matrices

We can express the problem in matrix form $A\mathbf{w} = \mathbf{y}$, where:

- A is the matrix of independent variables (features),
- $\mathbf{w}$ is the vector of weights (coefficients),
- $\mathbf{y}$ is the vector of prices (dependent variable).



Linear Models



- Linear models play a crucial role in solving engineering problems, where they are used to represent and approximate real-world systems and processes. In linear algebra, a linear model refers to an equation or system of equations that describe a relationship between variables using linear equations. Linear models are extensively applied in various fields of business, management and engineering, such as mechanical, electrical, civil, and systems engineering, to simplify and solve complex problems.

Introduction

- **Linear Models**
 - Practical problems in various fields can be represented using systems of linear equations.
 - These can be presented as a mathematical linear model in matrix form.
 - The matrix approach is beneficial as they can adapted to computer calculation to solve complex linear problems.
 - Matrix solution to linear models can be applied to various fields such as engineering, economics, ecology, etc.

The formula for the Cambridge Diet, a popular diet in the 1980s, was based on years of research. A team of scientists headed by Dr. Alan H. Howard developed this diet at Cambridge University after more than eight years of clinical work with obese patients. The very low-calorie powdered formula diet combines a precise balance of carbohydrate, high-quality protein, and fat, together with vitamins, minerals, trace elements, and electrolytes. Millions of persons have used the diet to achieve rapid and substantial weight loss.

Nutrition Model

- **Linear Model for construction a nutritious weight loss diet**
- Background Information:

Nutrition Model

Linear Model for construction a nutritious weight loss diet

Problem Statement:

To achieve the desired amounts and proportions of nutrients, Dr. Howard had to incorporate a large variety of foodstuffs in the diet. Each foodstuff supplied several of the required ingredients, but not in the correct proportions. For instance, nonfat milk was a major source of protein but contained too much calcium. So soy flour was used for part of the protein because soy flour contains little calcium. However, soy flour contains proportionally too much fat, so whey was added since it supplies less fat in relation to calcium. Unfortunately, whey contains too much carbohydrate. . . .

Nutrition Model

Linear Model for construction a nutritious weight loss diet

Problem: The problem is to , find some combination of nonfat milk, soy flour, and whey to provide the exact amounts of protein, carbohydrate, and fat supplied by the diet in one day

Amounts (g) Supplied per 100 g of Ingredient				Amounts (g) Supplied by Cambridge Diet in One Day
Nutrient	Nonfat milk	Soy flour	Whey	
Protein	36	51	13	33
Carbohydrate	52	34	74	45
Fat	0	7	1.1	3

Nutrition Model

Linear Model for construction a nutritious weight loss diet

Formulation:

The diet construction problem leads to the *linear* equation because the amount of nutrients supplied by each foodstuff can be written as a scalar multiple of a vector

	x_1	x_2	x_3
Nutrient	Nonfat milk	Soy flour	Whey
Protein	36	51	13
Carbohydrate	52	34	74
Fat	0	7	1.1

$\downarrow$
 $\mathbf{a}_1$

$\downarrow$
 $\mathbf{a}_2$

$\downarrow$
 $\mathbf{a}_3$

$$\left\{ \begin{array}{l} \text{Scalar} \\ x_1 \text{ units of} \\ \text{nonfat milk} \end{array} \right\} \cdot \left\{ \begin{array}{l} \text{Vector} \\ \text{nutrients per unit} \\ \text{of nonfat milk} \end{array} \right\} = x_1 \mathbf{a}_1$$

where $\mathbf{a}_1$ is the first column

let $\mathbf{b}$ be the vector that lists the total nutrients required

$$\begin{pmatrix} 33 \\ 45 \\ 3 \end{pmatrix}$$

Nutrition Model

Linear Model for construction a nutritious weight loss diet

Matrix Solution:

We can use the matrix equation $A\mathbf{x} = \mathbf{b}$ for solution.

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b}$$

The augmented matrix can be written as below:

$$\begin{bmatrix} 36 & 51 & 13 & 33 \\ 52 & 34 & 74 & 45 \\ 0 & 7 & 1.1 & 3 \end{bmatrix}$$

Nutrition Model

Linear Model for construction a nutritious weight loss diet

Matrix Solution:

Row reduction of the augmented matrix for the corresponding system of equations shows that

$$\begin{bmatrix} 36 & 51 & 13 & 33 \\ 52 & 34 & 74 & 45 \\ 0 & 7 & 1.1 & 3 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 0 & 0 & .277 \\ 0 & 1 & 0 & .392 \\ 0 & 0 & 1 & .233 \end{bmatrix}$$

To three significant digits, the diet requires .277 units of nonfat milk, .392 units of soy flour, and .233 units of whey in order to provide the desired amounts of protein, carbohydrate, and fat. ■